

Discrimination and Regional Disparities in Italy's Gender Wage Gap: An OLS and Blinder-Oaxaca Decomposition Analysis

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1 Introduction

The gender pay gap has remained a significant issue and has been a concern in labor economics across various economies. In Italy, there is a persistent issue that reflects deeper economic and social inequalities between men and women across various sectors of the labor market. The gender wage gap is defined as the difference in earnings between male and female employees. According to recent studies, the gender wage gap in Italy has persisted in a concerning manner, which estimates the difference between 15 and 20%, signifying that women earn significantly less than their male counterparts (Piazzalunga, 2015). Another critical aspect of the Italian gender pay gap is its regional difference with researches show significant disparities across different regions, influenced by factors such as local labor market conditions, firm types, and cultural attitudes towards gender roles (Storm, 2023).

In terms of methodology, many studies have utilized the Oaxaca-Blinder decomposition technique to analyze the gender wage gap. For example, Destefanis et al. (2023) utilize this method to explore the unexplained components of the wage gap in the Italian context, emphasizing the elasticity of earnings concerning work hours across different occupations. A similar study, Piazzalunga (2017) has used this technique to study the gender wage gap among college graduate in Italy.

This research builds upon this existing body of literature by employing a comprehensive regional analysis of the gender pay gap in Italy, utilizing the Blinder-Oaxaca decomposition method and microeconomic data from the Survey of Household Income and Wealth (SHIW) to observe overall national trends while dissecting the disparities that exist at the regional level. By doing so, this study aspires to contribute

meaningful insights into the pervasive issue of gender pay inequality, ultimately guiding policy recommendations tailored to specific regional contexts.

The study first applies a simple OLS regression on the log wage with only a gender dummy variable to find that, on average, Italian men earned 22% more than the women. Then, the study will be expanded to include additional productivity and characteristic controls. The results show that, after controlling for all observable variables, being a women still associates with having a 15.4% lower wage. To have additional insight into this gap, the study utilizes the Blinder-Oaxaca Decomposition to show that the majority (56.8%) is due to an unexplained component. This result signals that gender discrimination is the main driver of the gender wage gap in Italy, especially the difference between how men and women are being rewarded for their experience. Lastly, the study inspect this gap for each regions in Italy and found that this discriminatory effect is the strongest in the southern and central region of Italy and statistically insignificant in the northern region.

2 Data

2.1 Dataset

The Survey of Household Income and Wealth (SHIW) conducted by the Bank of Italy is this paper’s primary source of microeconomic data on household wealth in Italy. Since the 1960s, the survey has gathered data on the incomes, consumption, wealth, payment instruments, the characteristics of residential dwellings, form of insurance, and the sociodemographic characteristics of household members (gender, level of education, work, social origin, etc.) in an anonymous form. The sample in the most recent surveys comprises about 7,000 households (approximately 16,000 individuals), distributed over about 300 Italian municipalities, and after cleaning the data, we end up using around 4,000 observations for each wave of data.

For this study, we focus on wage earners and analyze gender disparities in earnings. The dependent variable of interest is *net wage*, while the key independent variables included are *gender*, *education*, *age/experience*, *employment type*, *industry*, and *contract type*, with a few other controls.

Despite the considerable attention paid to strategies aimed at gaining the trust of respondent households, the survey is affected by problems of selection bias, in other words, by lower levels of participation of wealthier households (D’Alessio and Faiella, 2002) and by under-reporting, i.e., statements regarding income or wealth that are not entirely truthful (D’Alessio and Cannari, 1990). Despite these problems, surveys have proven to be an irreplaceable instrument of economic analysis, as testified by

the more than 500 scientific works produced by more than 350 Italian and foreign authors listed in the bibliography of works based on survey data (Biancotti and D'Alessio, 2007).

2.2 Descriptive statistic

To study gender wage differences, Table 1 reports the summary statistics of the covariates used to model wage distributions computed on the subsample of full-time workers and disaggregated by gender.

Table 1: Summary Statistics for Workers by Gender

	Male		Female	
	Mean	S.D	Mean	S.D
Income	26247.01	29607.09	19864.76	16393.39
Age	46.35	12.57	47.83	11.22
Education	5.02	1.56	5.49	1.49
Married	0.34	0.47	0.25	0.43
Number of Children	1.52	0.73	1.60	0.71
Weekly Hours	38.81	8.24	33.71	9.98
Part Time	0.08	0.26	0.25	0.43
Occupation				
Blue collar	0.43	0.50	0.25	0.43
White collar	0.37	0.48	0.62	0.49
Manager	0.20	0.40	0.14	0.34
Graphical Area				
North	0.44	0.50	0.50	0.50
Centre	0.22	0.41	0.23	0.42
South	0.34	0.47	0.27	0.44

There is a substantial gender pay gap, with men earning, on average, around 30% more than women (€26,247 vs €19,865). The wage gap comes with a larger standard deviation for men, suggesting that men have a wider range of earnings, possibly due to the sticky floor pattern that mainly applies to women workers.

Women are slightly older on average (47.83 years) compared to men (46.35 years), while our age variable only includes observations with above 17 years of age, suggesting that women on average have a higher working experience. At the same

time, women have a higher mean education level (5.49 vs. 5.02). Despite having more experience and a higher educational level, they earn less than men. This reality challenges the assumption that higher education and experience should lead to higher wages, suggesting that other factors such as occupational segregation and discrimination may be at play.

Men are more likely to be in blue-collar jobs (43% vs. 25%) and managerial roles (20% vs. 14%), while women are more likely to be in white-collar jobs (62% vs. 37%). This is evidence for occupational segregation, with men dominating higher-paying blue collar and managerial roles, while women are concentrated in white collar jobs and have a higher chance of working a part-time job (0.25 vs. 0.08), which may offer lower wages. This could be due to our sample only studying women who have a job, leaving out a big proportion of women not working or housewives.

Additionally, women have a slightly higher mean number of children (1.60 vs. 1.52) which could reduce working hours or career interruptions (33.71 weekly hours worked for women vs. 38.85 hours for men). The difference in working hours could partially explain the wage gap. However, even after accounting for hours worked, the pay gap persists.

These findings underscore the need for policies that address occupational segregation, support work-life balance, and promote wage transparency to reduce the gender pay gap.

3 Empirical Model

3.1 OLS Estimation of the Gender Wage Gap

The study begins by analyzing the effect of gender on the log of the net wage received using a simple Ordinary Least Squares (OLS) method. The log of Net wage will simplify the interpretation of the model by allowing us to see the percentage effects of each independent variable on wage, instead of having to conceptualize the significance of varying euro amounts. The model is as follows:

$$\ln(wage_i) = \beta_0 + \beta_1 Female_i + \epsilon_i \quad (1)$$

where *wage* is Net wage, $\ln wage$ is defined as the natural log of Net wage. *Female_i* is the dummy variable for an individual to be female.

By only including a dummy variable for gender, we can see the raw percentage difference in wages between men and women in Italy. Our coefficient is expected to be negative and significant, consistent with the reality that women generally earn

less than men in the workforce. However, this model assumes that men and women are the same in all aspects, such as education and experience, which is untrue as shown in our descriptive statistics. Thus, it will suffer from omitted variable bias and cannot distinguish the discrimination factor.

The study's second model will be expanded to include productivity and characteristic elements that affect income as controls to isolate the discrimination effect. The expanded model is as follows:

$$\begin{aligned} \ln(wage_i) = & \beta_0 + \beta_1 Female_i + \beta_2 Education_i + \beta_3 Experience_i + \beta_4 Experience_i^2 \\ & + \beta_5 WeeklyHours_i + \beta_6 Partime_i + \beta_7 BlueCollar_i + \beta_8 WhiteCollar_i \\ & + \beta_9 Manager_i + \beta_{10} North_i + \beta_{11} Centre_i + \beta_{12} South_i + \beta_{13} Married_i \\ & + \beta_{14} Children_i + \varepsilon_i \end{aligned} \tag{2}$$

where $Education_i$ is an ordinal variable from 1 to 8, representing lowest to highest level of education. $Experience_i$ is a continuous variable which uses age as an indicator since our data only include working ages, and we also include a quadratic term $experience_i^2$ to capture the nonlinear effect of experience, as experience usually has a diminishing return on wages. Our study will include $Partime_i$ as a dummy for part-time workers. The occupation will be divided into three dummy variables: the blue collar worker, the white collar worker, and the manager. The same is true for the difference in geographical regions with three dummy variables for the observation in the north, center, or south of Italy. $Married_i$ is a dummy variable to indicate whether or not to be married and $Children_i$ is a continuous variable indicating the number of children.

The expectation is that the inclusion of individual components for characteristic and productivity will increase the model's explanatory power. $Education_i$ and $Experience_i$ are expected to be positive and significant, since investing in human capital signals greater productivity and contributes to earning a higher wage. The model uses $WeeklyHours_i$ as control for labor supply. An increase in weekly hours worked should contribute to a higher total wage. Being married and having a large family is expected to contribute negatively to total wage. By controlling for all the factors that largely explained the changes and differences in wage, the coefficient of interest β_1 can effectively show the gender wage gap that is "unexplained", or can be interpreted as gender-based discrimination.

3.2 Blinder-Oaxaca Decomposition

The OLS regression can allow the study to estimate the gender wage gap, however, it does not distinguish whether the gap is due to differences in characteristics (such as education and experience) or differences in how these characteristics are rewarded in the labor market. To overcome this limitation, we use the Blinder-Oaxaca decomposition.

The Blinder-Oaxaca decomposition offers an analytical framework to differentiate the factors contributing to wage gaps, particularly between gender and racial groups. Introduced by economists Alan Blinder (1973) and Ronald Oaxaca (1973), this decomposition provides valuable insights by allowing researchers to isolate the contributions of observable characteristics from unobservable discrimination present in labor markets, in our case, between working men and women in Italy. The method decomposes the total wage gap into two components: an **explained** part due to differences in observed characteristics (e.g., education, experience, occupation, etc.) and an **unexplained** part, which may reflect discrimination or unobserved factors.

3.2.1 Methodology

The wage equations for men and women can be written as:

$$\ln w^M = X^M \beta^M + \varepsilon^M \quad (3)$$

$$\ln w^F = X^F \beta^F + \varepsilon^F \quad (4)$$

where:

- w^M, w^F represent men's and women's wages, respectively.
- X^M, X^F are vectors of observed characteristics (e.g., education, experience, occupation).
- β^M, β^F are the estimated coefficients for each group.
- $\varepsilon^M, \varepsilon^F$ are the error terms.

Taking expectations and rearranging terms, the mean wage difference can be decomposed as follows:

$$\bar{w}^M - \bar{w}^F = (\bar{X}^M - \bar{X}^F) \hat{\beta} + \bar{X}^F (\hat{\beta}^M - \hat{\beta}^F) \quad (5)$$

The first term $(\bar{X}^M - \bar{X}^F)\hat{\beta}$ is the **explained** component, which corresponds to the wage differences due to different characteristics. This component reflects differences in education, experience, occupation, and other observable factors. If men and women had the same characteristics, this part of the gap would disappear.

The second term $\bar{X}^F(\hat{\beta}^M - \hat{\beta}^F)$ is the **unexplained** component, capturing differences in the return to characteristics. This part remains even after controlling for characteristics and often labeled as “discrimination”, as there should be no market reason to have different wage returns to the same characteristics for men and women. However, the unexplained component may include unobserved characteristics effect.

3.2.2 Assumptions

The Blinder-Oaxaca decomposition method relies on several key assumptions to ensure that its estimates accurately reflect wage differences due to observable characteristics and unexplained factors.

Firstly, the Common Support Assumption indicates that the set of variables influencing wage levels must be the same across groups (e.g., men and women). This ensures that the comparison is meaningful and that there is overlap in the characteristics of both groups.

Secondly, the Conditional Mean Independence Assumption assumes that the distribution of unobservable characteristics should be independent of group membership when conditioned on observable traits. This allows the decomposition to separate the effect of differences in characteristics (composition effect) from the differences in how those characteristics are rewarded (returns to observables).

Lastly is the Invariance of Conditional Distributions: The conditional wage distribution of the reference group (e.g., men) should remain valid if the other group (e.g., women) were paid under the same wage structure. This ensures that the counterfactual wage estimation used in the decomposition is meaningful.

If these assumptions hold, the explained component of the decomposition represents only the differences in observable characteristics, while the unexplained component reflects differences in how those characteristics are valued in the labor market, which may indicate discrimination or other unobserved factors.

4 Result

4.1 OLS regressions Results

The results of the Ordinary Least Squares (OLS) regressions are reported in Table 2. In the first model, the coefficient for the female dummy variable is -0.220 and statistically significant at the 1% level. This indicates that women earn approximately 22% less than men in Italy when no other factors are controlled for. This raw wage gap is consistent with the descriptive statistics presented earlier, which showed that men earn significantly more than women on average. The second model accounts for a range of control variables that can help better explain this income gap, including age, education, weekly hours worked, part-time status, occupation type, geographical region, marital status, and number of children. Inclusion of these variables increases the explanatory power of the model, as shown by the increase in the R-squared value from 0.025 in the first model to 0.539 in the second model. This would seem to indicate that Model 2 is preferable to Model 1. When controls are included, Model 2's coefficient for female dummy variable decreases to -0.154 but still remains statistically significant at the 1% level. This means that even after controlling for various productivity and characteristic factors, women still earn approximately 15.4% less than men. This residual wage gap suggests that there is an unexplained portion of the gender wage gap, which can be due to factors such as discrimination.

The control variables also offer important information on the wage determinants in Italy. Age, which represents work experience, increases wages by 3.08% for each year of experience, though this effect decreases overtime (-0.00024 for age squared), reflecting the diminishing returns to experience. Each higher level of education corresponds to a 7.17% increase in income. An additional weekly hour worked also increases income by 1.87%, while part-time work significantly reduces earnings by -36.1%, highlighting the economic penalty associated with part-time employment, which is more common among women. Blue-collar workers earn significantly less than managers, by 70.9%, as do white-collar workers (46.1%). This shows the presence of occupational segregation, where managers, who are more likely to be men, earn higher wages compared to both blue-collar and white-collar workers, with women disproportionately concentrated in lower-paying white-collar roles. Interestingly, marital status and the number of children do not have a significant impact on wages in this model. This suggests that the number of children does not play a major role in explaining the gender wage gap in Italy, at least within the context of the variables controlled for in this analysis.

Table 2: OLS Regression Results

VARIABLES	(1) Log Income	(2) Log Income
Female	-0.220*** (0.0219)	-0.154*** (0.0174)
Education		0.0717*** (0.00664)
Age		0.0308*** (0.00659)
Age ²		-0.000243*** (7.27e-05)
Weekly Hours		0.0187*** (0.00154)
Part-time		-0.361*** (0.0301)
Blue-collar		-0.709*** (0.0328)
White-collar		-0.461*** (0.0262)
Manager		-
North		0.183*** (0.0175)
Centre		0.103*** (0.0214)
South		-
Married		-0.122*** (0.0203)
Children		-0.00220 (0.00962)
Constant	9.905*** (0.0155)	8.395*** (0.169)
Observations	3,969	3,969
R-squared	0.025	0.539

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

4.2 Blinder-Oaxaca Decomposition Result

The results of the Blinder-Oaxaca decomposition are summarized in Table 3. Consistent with previous findings in our OLS regression analysis, the raw gender income gap is 22% with men earning more than women. With Blinder-Oaxaca decomposition, we can see that the explained component only accounts for 18.5% and the unexplained component accounts for the majority 56.8% of the gap. And the interaction term contributes to about 24.6% of the gap. This large unexplained component shows that discrimination and unobserved factors significantly affect wage disparities.

The endowment (explained) component contributed 0.0408, around only 18.5% of the gap, suggesting that the difference in characteristics between men and women is only responsible for a minority of the income difference. This component suggests that men work more hours on average and are more likely to hold higher-paying managerial and blue-collar roles, contributing to the wage gap. However, women have slightly more experience and are more educated, which helps partly reduce the gap.

The coefficient component (unexplained) represents the differences in how men and women are being rewarded for the same characteristics. This account for 0.125, equal to 56.8% of the raw income gap and it is statistically significant. This suggests that discrimination plays a major role in the gap, however, individual variables within this component, such as education, occupation, hours worked, etc., are mainly statistically insignificant for our sample. The exception is the return on experience. The statistically significant negative coefficient for the linear term for experience means that women are rewarded less than men for each additional year of experience. The quadratic term has a positive effect, showing that women's wage grows faster than men at a very high level, but this effect is small and only partially offsets the negative return to experience. The fact that only experience is statistically significant in the coefficients component suggests that differences in returns to experience are the primary driver of the unexplained portion of the gender wage gap in our sample.

Finally, the interaction component, which accounts for 0.0541 (24.6% of the total gap), captures the interaction between differences in characteristics and differences in returns to those characteristics. This component is also statistically significant, highlighting the complex interaction between observable differences and how they are rewarded in the labor market.

Table 3: Decomposition of Differences in Log Income

VARIABLES	(1) Overall	(2) Endowments	(3) Coefficients	(4) Interaction
Age		-0.0663*** (0.0230)	-1.133* (0.644)	0.0354 (0.0221)
Age^2		0.0435** (0.0183)	0.666* (0.354)	-0.0300 (0.0185)
Education		-0.0348*** (0.00554)	-0.0261 (0.0731)	0.00226 (0.00633)
Weekly Hours		0.0989*** (0.0120)	-0.0378 (0.103)	-0.00575 (0.0157)
Part-time		0.0557*** (0.00697)	-0.0296 (0.0180)	0.0207 (0.0127)
Married		-0.00876*** (0.00292)	-0.0111 (0.0101)	-0.00407 (0.00375)
Children		-0.000223 (0.000690)	-0.00386 (0.00884)	0.000431 (0.00101)
North		-0.00535** (0.00221)	-0.00988 (0.0204)	0.00120 (0.00250)
South		-0.00612** (0.00263)	-0.00979 (0.0116)	-0.00260 (0.00312)
Blue-collar		-0.143*** (0.0147)	0.0236 (0.0167)	0.0177 (0.0126)
White-collar		0.108*** (0.0114)	-0.0460 (0.0324)	0.0187 (0.0132)
Male	9.905*** (0.0156)			
Female	9.685*** (0.0155)			
Difference	0.220*** (0.0220)			
Endowments	0.0408** (0.0195)			
Coefficients	0.125*** (0.0214)			
Interaction	0.0541*** (0.0177)			
Constant			0.743** (0.359)	
Observations	3,969	3,969	3,969	3,969

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

5 Gender Pay Gap by Regions in Italy

In the previous section, we used the Blinder-Oaxaca decomposition to study the gender wage disparity for the sample of the general population of Italy and found that the majority of the difference was due to discrimination or unobserved factors. In this section, we repeat the same procedure for each region in Italy to see whether the discriminatory effect differs geographically. Since the socioeconomic situation in Italy is vastly different between the regions, this could contribute to a difference in the wage gap. The northern region is considered the most economically developed with a robust industrial base, such as the city of Milano, Genova, etc., a high employment rate, and great access to advanced education and opportunities (ISTAT, 2020). The centre region, while also relatively prosperous, has a mixed economic structure with a slower growth rate compared to the north (ISTAT, 2020). The southern region lags behind in economic development, with higher unemployment rates and lower wages, and is heavily dependent on agriculture and tourism (OECD, 2020).

Table 4 reports the Oaxaca decompositions for each of the three regions in Italy: North, Centre, and South. The results show that the magnitude of the gender pay gap is not uniform across the country, with the central region having the largest overall gender pay gap at 29.6%, followed by the northern region at 26.7%, and the southern region at 13.3%.

Table 4: Decomposition of Gender Pay Gap by Region

Region	Difference	Endowments	Coefficients	Interaction
Northern Region	0.267*** (0.0320)	0.100*** (0.0267)	0.062 (0.0364)	0.105*** (0.0317)
Central Region	0.296*** (0.0505)	0.130*** (0.0447)	0.151*** (0.0434)	0.015 (0.0351)
Southern Region	0.133*** (0.0378)	-0.084** (0.0367)	0.157*** (0.0313)	0.061** (0.0277)

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

In the central region of Italy, men are paid, on average, 29.6% more than women. More than half (51%) of his gap is driven by the unexplained component, as indicated by the significant coefficient effect of 0.151. The other half is contributed by the endowment effect, which also plays a notable role, with a significant coefficient of

0.13. This indicates that women in the Central Region are not only disadvantaged by differences in observable characteristics but also by how these characteristics are rewarded.

While still having a substantial gender wage gap of 26.7%, the composition of this gap in the north is different. With significant contributions from endowments and the interaction term, interestingly, the unexplained component (coefficients) is not statistically significant. This means that gender discrimination does not significantly contribute to the pay gap in this region. This could be attributed to the more advanced socio-economic development in the North, which may lead to more standardized and transparent pay structures, reducing the impact of discriminatory practices in wage determination.

Lastly, the Southern Region, despite its economic challenges, shows the lowest overall pay gap of only 13.3%. In contrast with the other regions, the observable component reports a negative effect of -0.084. This suggests that women in this region generally have better characteristics and this contributes to lowering the pay gap. However, this effect is being offset by the discriminatory component of 0.157, the largest between the regions. This signals that women are the most disproportionately underpaid and wage discriminated in the south.

6 Conclusion

This study provides a comprehensive analysis of the gender wage gap in Italy using microeconomic data from the Survey of Household Income and Wealth (SHIW). Despite women, on average, having more working experience and being more educated in our sample, results show that a significant wage gap exists between men and women in Italy, with women earning, on average, 22% less than men. Using OLS regression, even after controlling for key factors such as education, experience, working hours, and occupational segregation, we still find that the gender wage gap remains at approximately 15.4%, suggesting that structural and discriminatory factors play a substantial role.

We then use the Blinder-Oaxaca decomposition to further show that the majority of the wage gap is unexplained by observable characteristics, indicating that there are significant discrimination that contribute to wage disparities in the whole sample. In particular, the majority of the discrimination effect comes from women receiving lower returns on experience, which may reflect persistent discrimination in career advancement and wage progression.

Additionally, we have done separate Blinder-Oaxaca decomposition for each region to show that the gender pay gap varies across Italy, with the largest overall wage

disparities found in the central region and the highest level of unexplained wage discrimination in the south. This result suggest there is a need to have different policy intervention for different regions due to distinct socio-economic conditions of each region.

These findings highlight the need for targeted policy interventions to reduce the gender wage gap. Policies promoting wage transparency, stronger anti-discrimination measures could help address the underlying causes of pay disparities. Additionally, there are more to be done to reduce occupational segregation and ensure equal career advancement opportunities, and to foster a more equitable labor market.

While this study sheds light on the gender wage gap in Italy, further research is needed to explore unobservable factors, such as negotiation dynamics, implicit biases, and employer practices. Addressing these issues is crucial to achieving wage equality and ensuring that labor market outcomes are determined by merit rather than gender.

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